The Youth Science Journal Of Iraq

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April Fool's Edition
Staff Edition

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A Challenge And A Note:

This edition of YSJOI is our Staff edition, where each of the staff picked out their choice of questions and it was answered. To keep the fun of the edition, the articles may or may not be misattributed to random people; and hence the challenge:
One of the articles of this issue is completely false; our own little April fool's joke. See if you can recognize it!

The answer will be posted in the next issue of YSJOI!

A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: What is the best heat conductor?

– *Safa Wissam*

D for diamond, right? Well, you might just be wrong this time! Diamond has been the best heat conductor for over 250 years... So, why is it not quite the correct answer?

Before the answer is revealed, let's ask another question: What makes a material good at conducting heat?

A substance's heat conductivity depends on its structure; if its internal structure allows energy to move from one atom to another with minimal resistance through carriers such as delocalized electrons in metals, then it's fit for the job.

And that is exactly what diamond is—well, not quite. Diamond is made up of carbon atoms linked by tight bonds in a 3D, tetrahedral structure. Carbon is also a light element, it can vibrate at very high frequencies, allowing heat to zip through the structure. Think of it like a tight guitar string versus a loose one—the tight string transmits energy more efficiently. And since diamond's structure is a repeating lattice, it allows heat to travel in a straight line without facing any obstacles. But, if diamond is made up of carbon, how can it have delocalized electrons? It doesn't!
Instead of freely-moving electrons, diamond has phonons; a single unit of traveling vibration. Due to diamond's tight, stiff bonds, heat waves (phonons) are able to zip through in an instant!
Recently, scientists at the University of Houston have proven that *Boron Arsenide* is better at conducting heat than diamond—reaching over 2100 W/mK at room temperature, exceeding the theoretically predicted value of just 1280 W/mK. While diamond stands between a range of 2000-2200 W/mK, boron arsenide is a huge competitor, not only physically, but economically as well. It does not require extreme temperatures nor pressures, making it a lot cheaper to produce. Both diamond and boron arsenide can be classified as thermal grades; they efficiently allow extreme heat dissipation in electronics, preventing them from melting under high power. But, here's the kick: boron arsenide is also a great semiconductor. Therefore, not only can it operate as a cooling system, it can also semi-conduct electricity.
In conclusion, diamond has its competitor, so it's not exactly the right or wrong answer *yet*—only time will tell.

Question II: What is the universe fated to be?

– *Ahmed Muhammed*

The Big Bang. The most widely accepted theory about what started this Universe. Since then, it has been expanding at increasing speeds. But how will it end?

There are three main theories: heat death, the Big Rip, and the Big Crush.

First of all, heat death. This is entropy, where all stars eventually stop burning, and all heat undergoes radiation, conduction, and convection, so that the whole world is in thermal equilibrium. In such a case, all energy will be balanced out, and no more reactions, energy transfer, or anything will happen. The Universe would have simply exhausted itself. In this case, the Universe is not destroyed; it just becomes one big beaker at a constant temperature. It's the boring ending.

On the other side, there is the Big Rip. This theory states that the Universe will grow, and continue to grow, accelerating, so that gravity eventually becomes too weak to hold it together, and the Universe rips itself to shreds. It starts with superclusters of galaxies getting ripped apart, then galaxies breaking down, then solar systems, eventually leading to the breakdown of individual atoms, and everything would get reduced to… nothing. Everything would cease to exist as the four fundamental forces are simply unable to do their job.

The third theory now presents itself: the Big Crush. In this ending, the opposite of the previous one happens.

As the Universe continues to expand, and as it expands violently, gravity gets itself together and pulls in the Universe, fighting against the expansion. Eventually, gravity wins, and the reverse of the Big Bang happens, transforming our Universe from an infinite expanse of free space to a hyper-energized dense cluster of energy, just like how it was when the Universe began. However, this is highly unlikely, as if it does happen, the Universe would simply expand again.

Now this begs the question: if the Big Crush theory is true, then is this the first iteration of the Universe?

Question III: What would happen if the sun disappeared?

– *Abdullah Haider*

The Sun is the largest object in our solar system, shining bright to warm us and give us light to see. A main reason that life on Earth exists. But what would happen if it disappeared? No explosion or any cosmic event - it just disappears.

Many things would happen quite quickly if the Sun disappeared. What first comes to mind is that Earth goes dark the instant our main light source is gone; however light does not travel instantly – it takes about 8 minutes to reach earth – so we would still have light for another 8 minutes. The same happens to our orbit; gravity also 'travels' at approximately the same speed of light so therefore we would spin around a seemingly imaginary point for a few minutes before being slung out of orbit at a tangent to float away into space - a rogue planet followed only by our moon.

What else happens? After the first few minutes we would already be lost in space, wandering aimlessly until we get caught by another gravitational field, but what next? Days pass by and global temperatures decrease, because there is no longer anything to maintain temperatures and no energy source besides the relatively small amount stored in Earths' core. This means we can no longer breathe without damaging our lungs by cold, dry air now present in our atmosphere, which gets thinner and thinner as it condenses due to the freezing temperatures. Not so good a start for all living beings when the Sun disappears.

The few people capable will retreat to underground bunkers if available and the rest will have to suffer this possible extinction-level event – that's how bad it will get. Even the few underground will eventually run out of resources and most of Earths creatures will be wiped out by a month of the Suns disappearance. Except for some organisms living under the high pressures at the bottom of the sea (this prevents the water from freezing so keeps them alive), these will probably be the only form of life present on Earth and are forced to remain there since everywhere other than these high-pressure regions would have frozen or become uninhabitable.

Question IV: How does the brain dream of things it has never experienced?

– Parwar Phicri

Dreams about falling from great heights are very common, even though many people have never experienced it in real life. This raises a question: how can the brain create something so realistic and intense without any memory of it?

One idea is that the brain does not need experience to create images of danger, but instead uses survival instincts. In this case, it is the fact that humans are naturally aware of gravity and balance. These ideas are deeply stored into our nervous system from the time we are babies, when we learn to walk, climb, and jump.

Another explanation involves the imagination. During sleep, the brain becomes highly creative; it mixes memories, emotions, and thoughts to form dreams. Even if someone has never fallen from a cliff or been stabbed, they may have seen similar situations in stories or movies, or heard others describe the feeling. The brain then builds on these small pieces to create an illusion.

Emotions also play a major role. Dreams of falling are often linked to anxiety, stress, or a sense of losing control in life. The falling sensation becomes a symbol rather than a real event; it reflects inner thoughts, not real experiences. In this way, the mind is not limited by what we have physically lived. It can combine instinct, imagination, and emotion to produce dreams that feel real and sometimes terrifying, even without real experience of the event itself.

This ability shows how powerful and complex the human brain actually is, as it can simulate whole experiences from nothing. Dreaming is not just a summary of what physical experiences you have had; instead, it shows your deeper nature, which many are unaware of. So in reality dreams of falling from heights show that the mind is not limited to experiences in the real world. Instead, it also takes use of imagination, and emotions to create realistic situations that might have not actually been lived through. Moreover these dreams are not just random events, but a meaningful reflection of our thoughts and feelings, truly showing us how complex the mind is.

Question V: How do black holes work?

– *Ali Ahmed Kayani*

Black holes are one of the most mysterious and fascinating objects in the universe. They are regions in space where gravity is so strong that nothing, not even light, can escape. This idea alone sounds almost impossible, yet scientists have proven that black holes are real and exist far beyond our planet.

Most black holes are formed when massive stars collapse at the end of their life cycle. As the star shrinks, its mass gets packed into an incredibly small space, creating a powerful gravitational pull. Anything that comes too close gets pulled in, crossing what scientists call the “event horizon,” a point of no return.

What makes black holes even more interesting is that we cannot see them directly. Since they do not emit light, they are invisible. However, scientists detect them by observing how nearby stars and gases behave. When matter is pulled into a black hole, it heats up and produces intense radiation, giving us clues about its presence.

Despite the progress in space science, many questions about black holes remain unanswered. What happens inside them? Do they lead to another part of the universe? Some theories even suggest the possibility of time distortion, where time slows down near a black hole. These ideas may sound like science fiction, but they are based on real scientific thinking.

At the current moment in time everyone’s guess is good as any other. Maybe there’s magic in a black hole, or maybe another universe. What’s certain is that we will learn more about them in our lifetimes! Maybe it will be one of us, the youth, who figures it out!

Black holes remind us that the universe is far more complex than we imagine. They challenge our understanding of physics and push scientists to explore deeper into space. For students like us, they show that science is not just about answers, but also about endless questions waiting to be explored.

Question VI: Why is the Moon’s density so low?

– *Yousif Muhammed*

We have all looked up at the night sky and wondered about the glowing orb that orbits our planet. For centuries, we were told it was a desolate rock, a silent witness to Earth’s history made of basalt and regolith. We grew up believing the Apollo missions brought back "moon rocks" that were heavy, solid, and ancient.

But does this actually make sense?

If the Moon were truly a solid sphere of rock, its gravitational pull would be significantly higher than what we observe today. Scientists have recently begun to question the "Lunar Density Paradox." When you look at the way the Moon "rings" like a bell when hit by meteorites—a phenomenon recorded by seismic sensors—it suggests that the interior isn't solid at all. It is porous. It is light. It is, essentially, a celestial ball of aerogel.

Aerogel, often called "frozen smoke," is the lightest solid known to man. It is 99.8% air and an incredible insulator. This explains why the Moon can withstand the extreme temperature swings of space without crumbling. But how did it get there?

New theories suggest that during the early formation of the solar system, high-pressure "quantum foam" solidified in the cold vacuum, trapping trace amounts of stardust within a silica-based lattice. This created a massive, low-density sphere. The "rocks" brought back by astronauts were actually just the thin, outer crust—a "scab" of space dust that settled on the aerogel core over billions of years.

This leads to an even more exciting realization: if the Moon is a giant ball of aerogel, it doesn't just orbit us; it floats on the solar winds! This is why the magnetic field of the Earth is so important; it keeps our "smoke balloon" from drifting away into the deep dark.

So next time you look up, don't see a heavy rock. See a delicate, frozen cloud of smoke, held in place by the delicate balance of the universe. It really is rather cool to think about, since it is a matter of this balance.

Acknowledgements

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We thank all the scientists who have worked thus far to find all about the world so we can learn about it, and everyone who has supported any member of the YSJOI team in furthering our mission.

Thank you to the rest of the YSJOI team (listed on the next page) for this incredible staff edition and April Fool's edition in the April 2026 issue of YSJOI!

Thanks also to the growing team of YSJOI,
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all of who had a hand in this incredible issue
and all the work that went into it:
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And we thank you, reader, for supporting our
mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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